MULTI-CAMPUS LAYER2 NETWORK FABRIC

William Dauchy - Gandi.net

Kernel Recipes 2015

GANDI.NET

DOMAIN NAMES

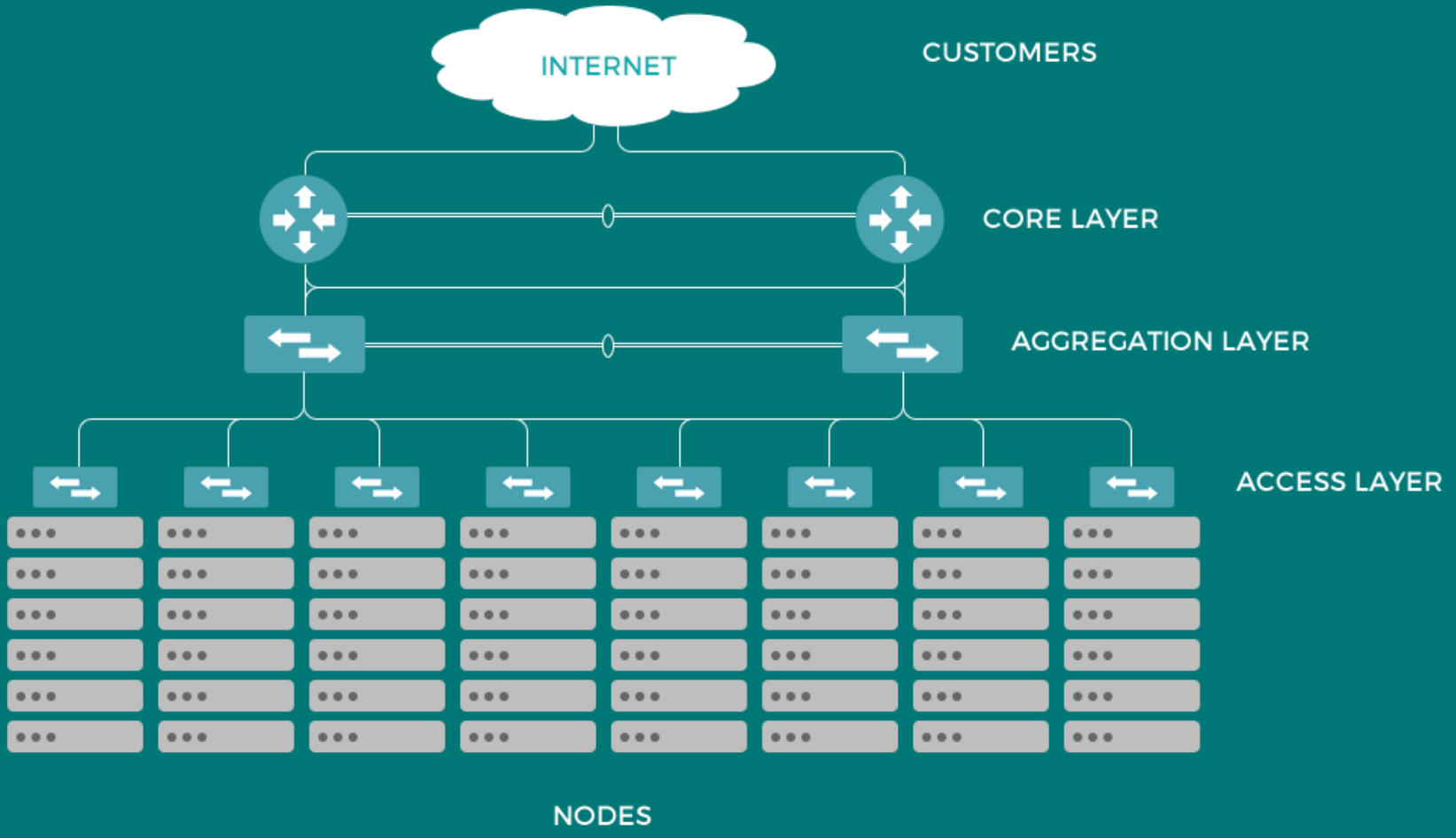
SSL

SIMPLE HOSTING

SERVER

KERNEL RECIPES 2013

- provide large scale multi-tenancy

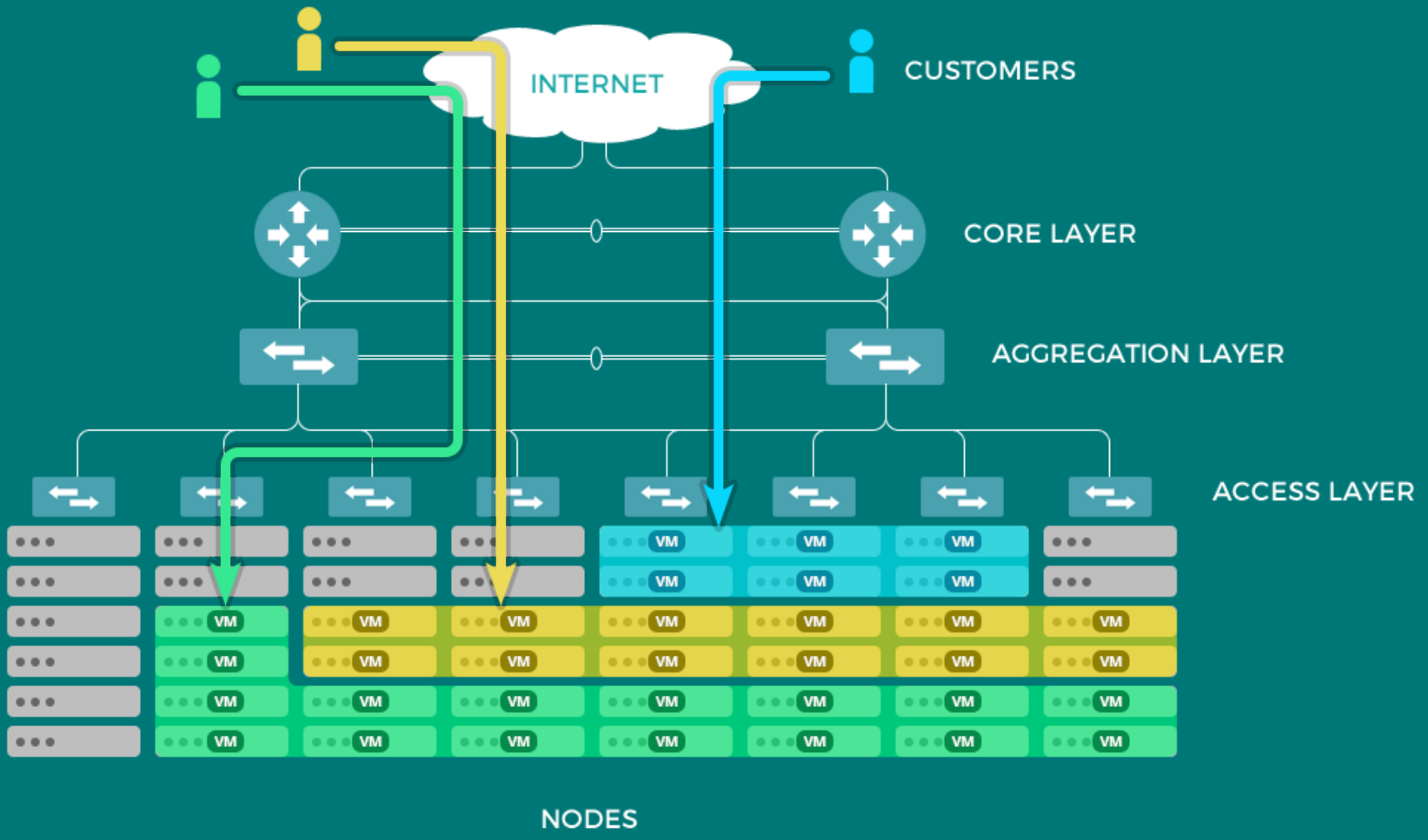

INTERNET
CUSTOMERS
CORE LAYER
AGGREGATION LAYER
ACCESS LAYER
NODES

KERNEL RECIPES 2013: REQUIREMENTS

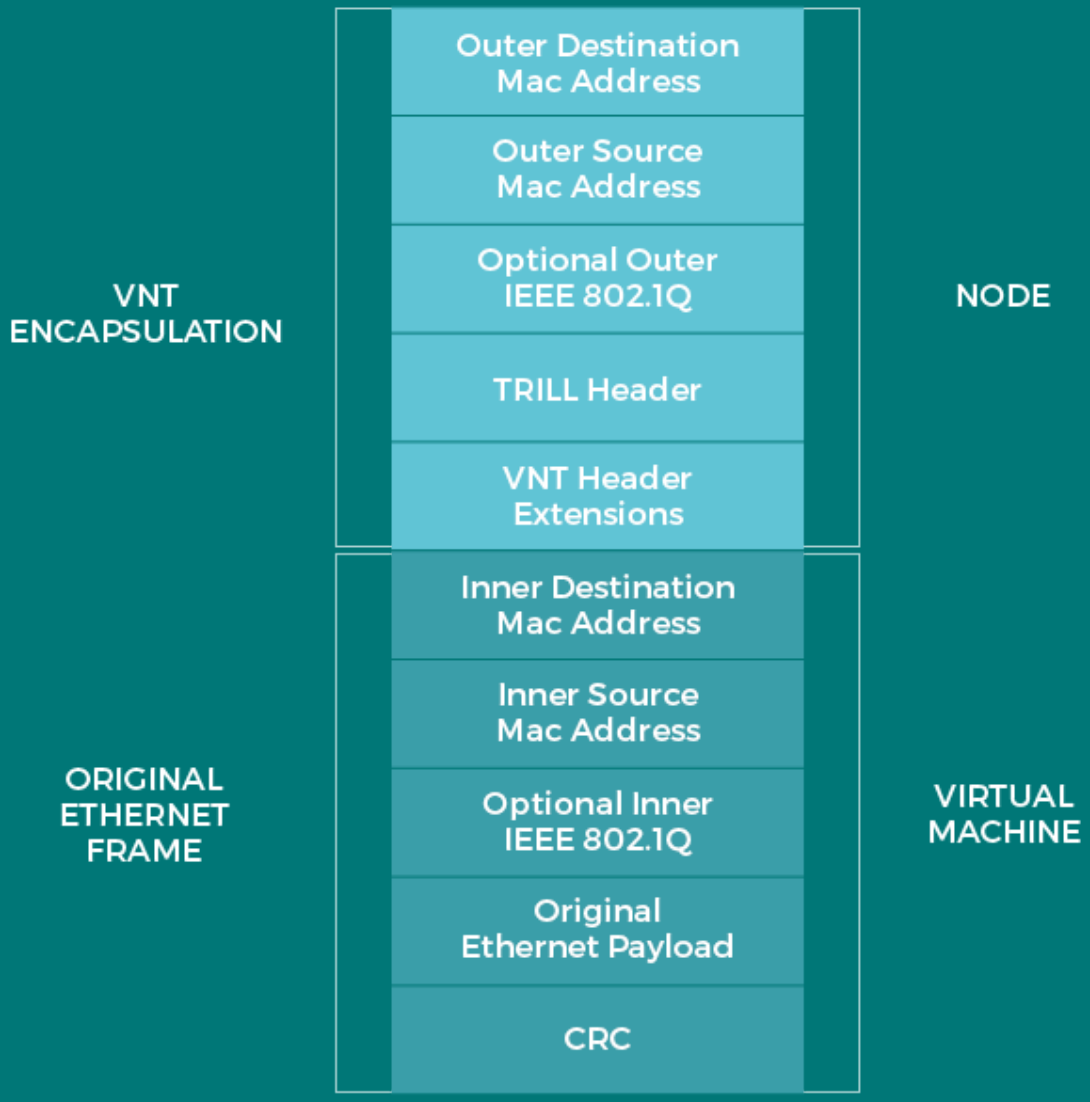
- Seamless VM mobility
- Easy management
- Layer 2 core scaling
- Fault resiliance
- VLAN scalability

LARGE SCALE MULTI-TENANCY

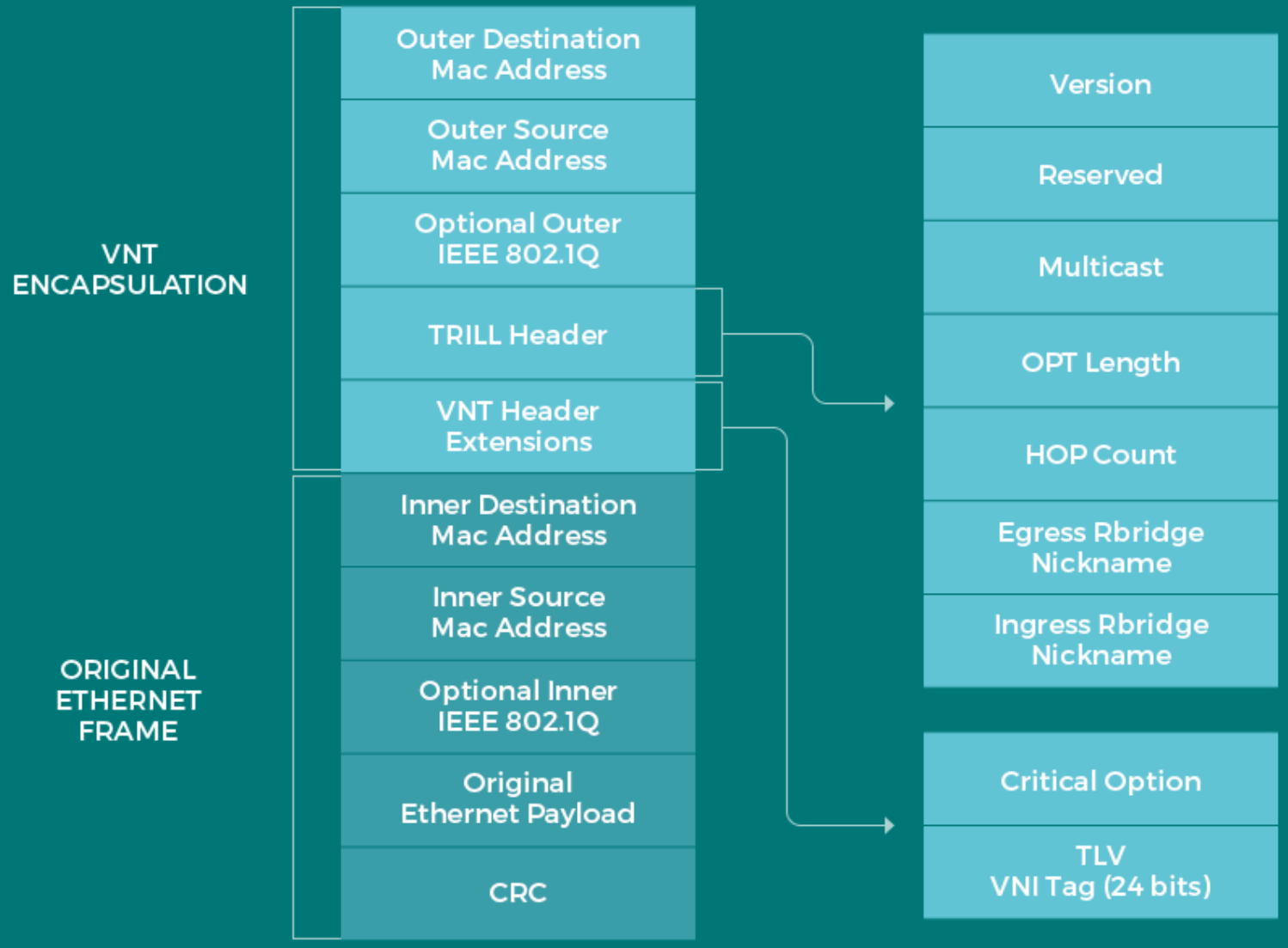
Multiples users using same resources



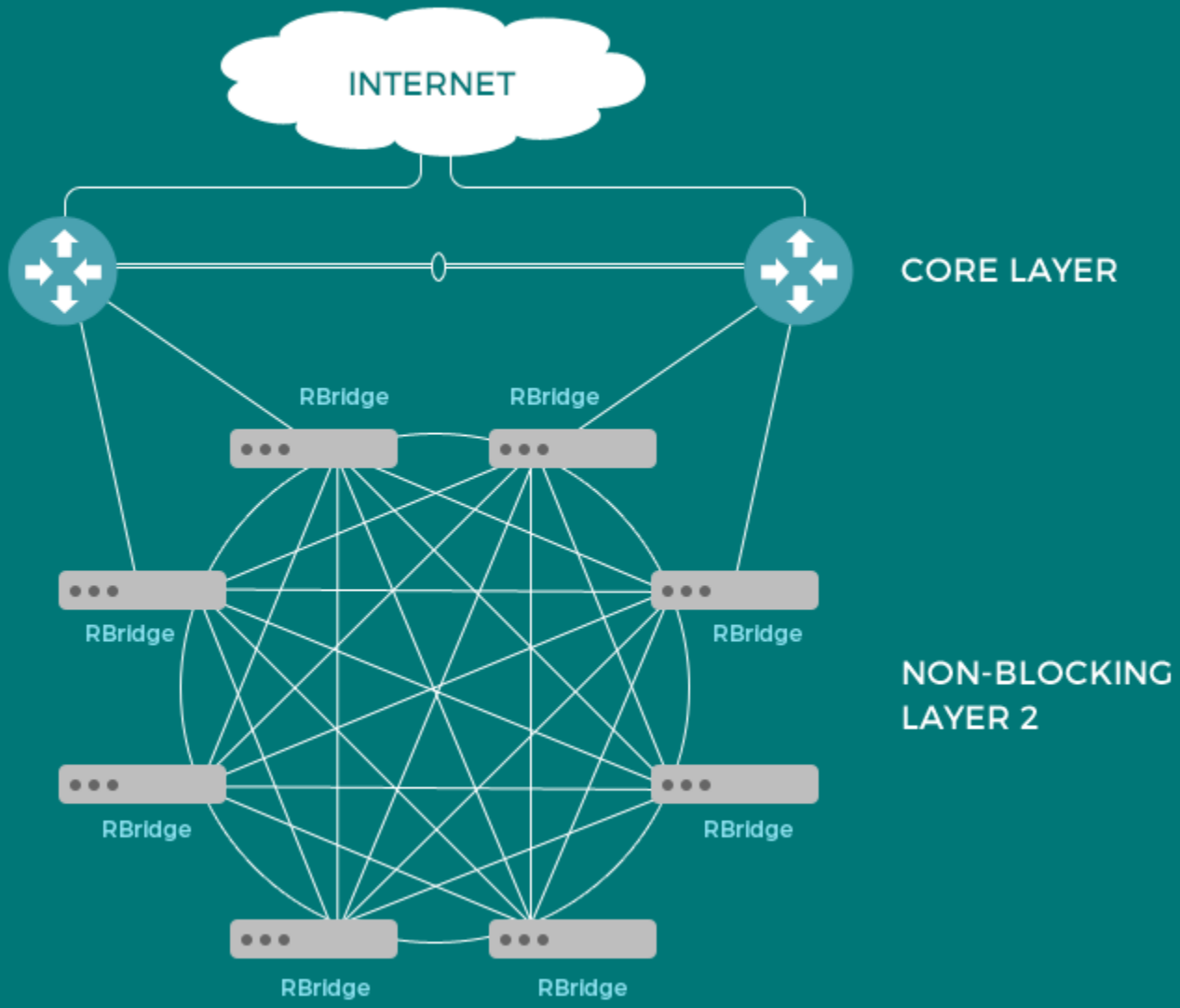
HEADER


VNT ENCAPSULATION
Outer Destination Mac Address
Outer Source Mac Address
Optional Outer IEEE 802.1Q
TRILL Header
VNT Header Extensions
NODE
ORIGINAL ETHERNET FRAME
Inner Destination Mac Address
Inner Source Mac Address
Optional Inner IEEE 802.1Q
Original Ethernet Payload
CRC
VIRTUAL MACHINE

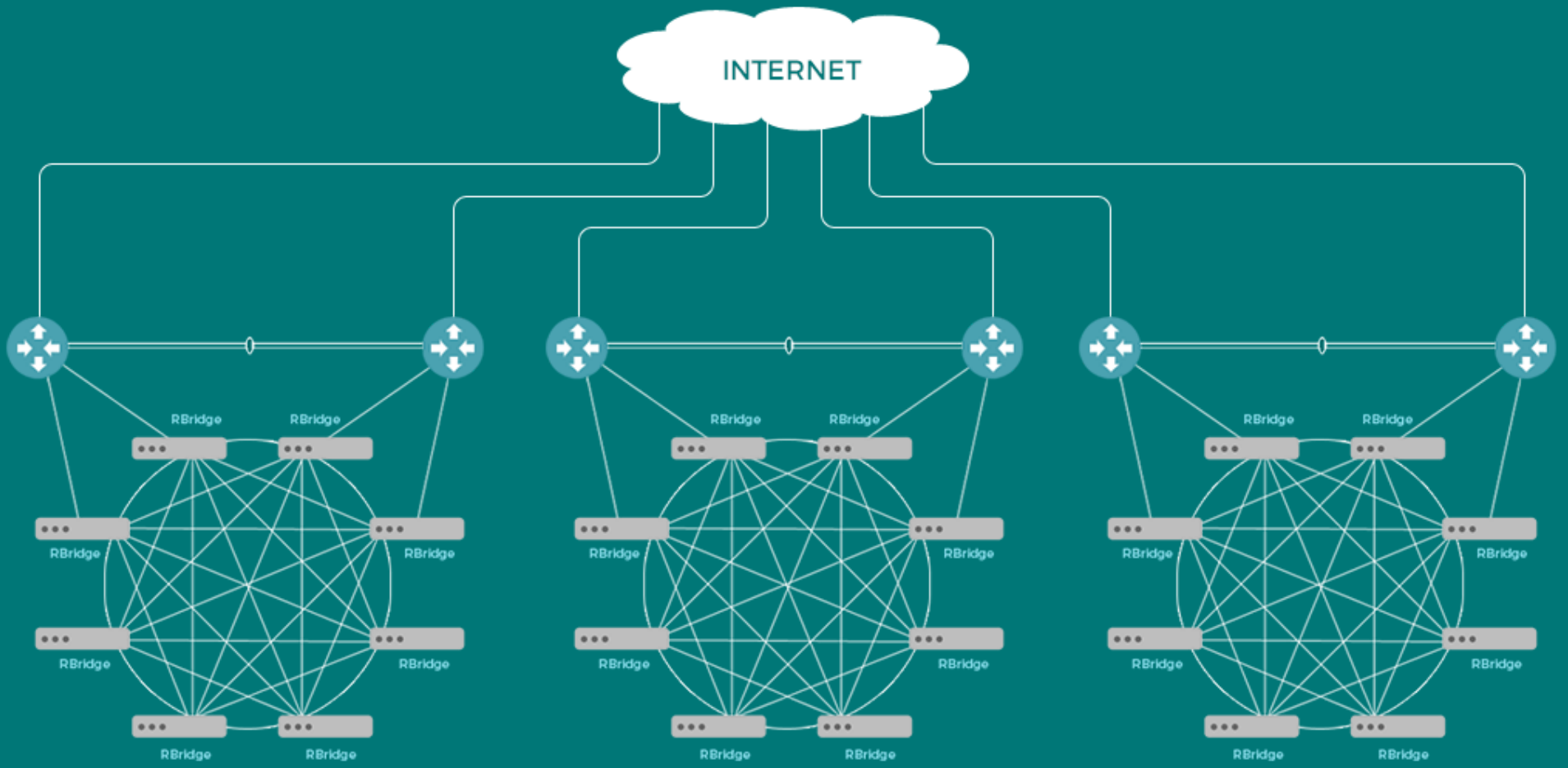
TRILL + VNT FRAME FORMAT



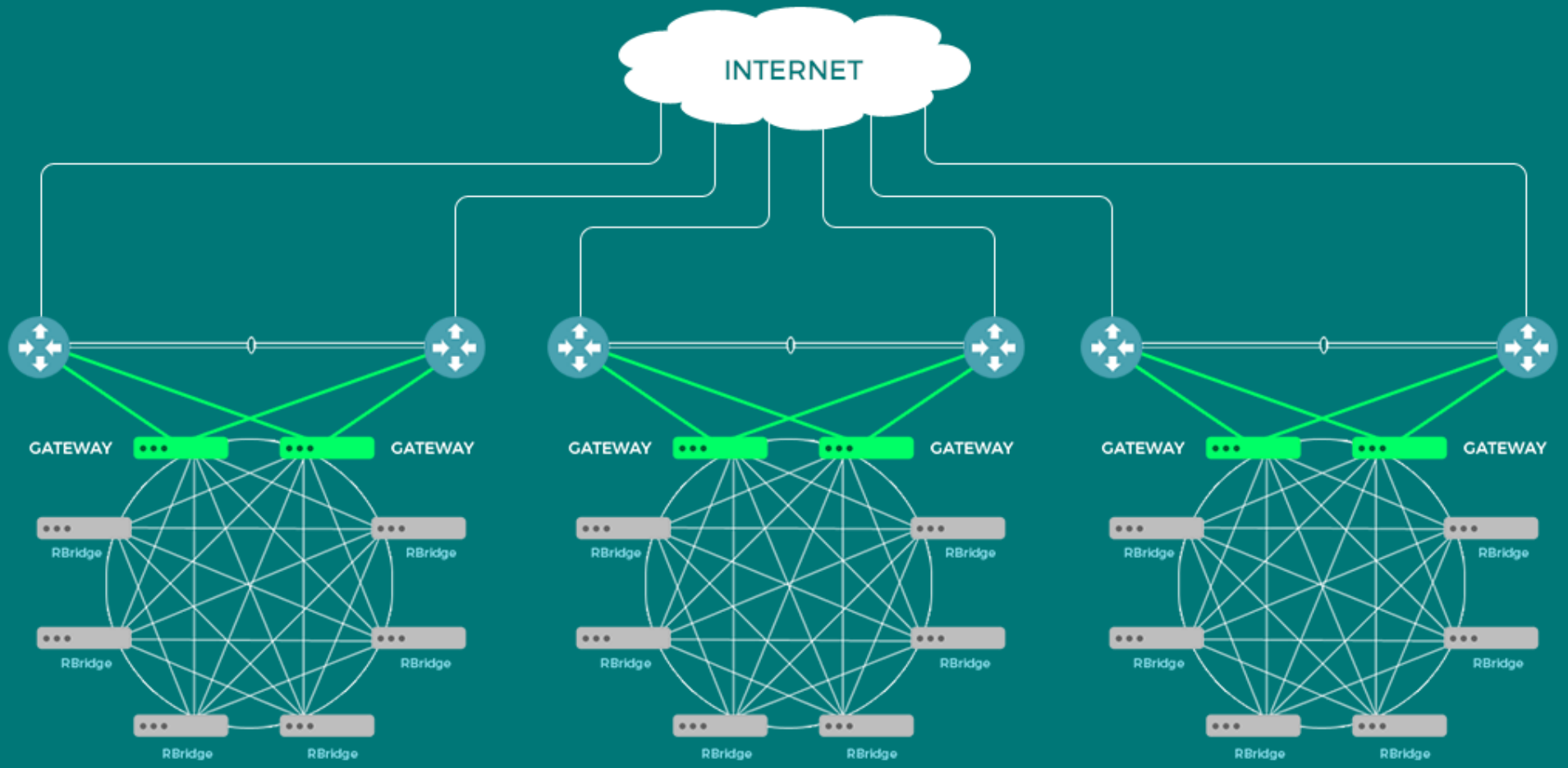
TRILL BASED DATA CENTER



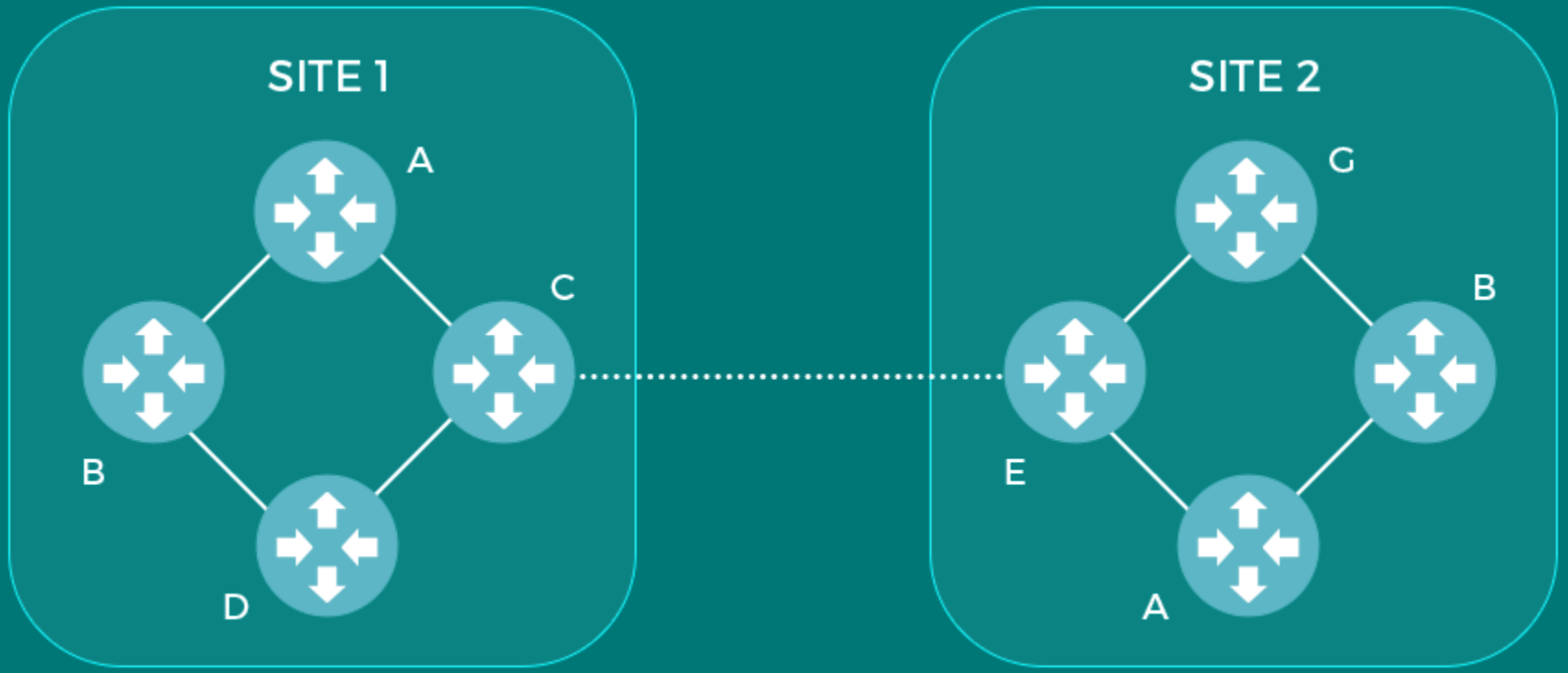
MULTI CAMPUS DATACENTER



MULTI CAMPUS TRILL BASED DATACENTER



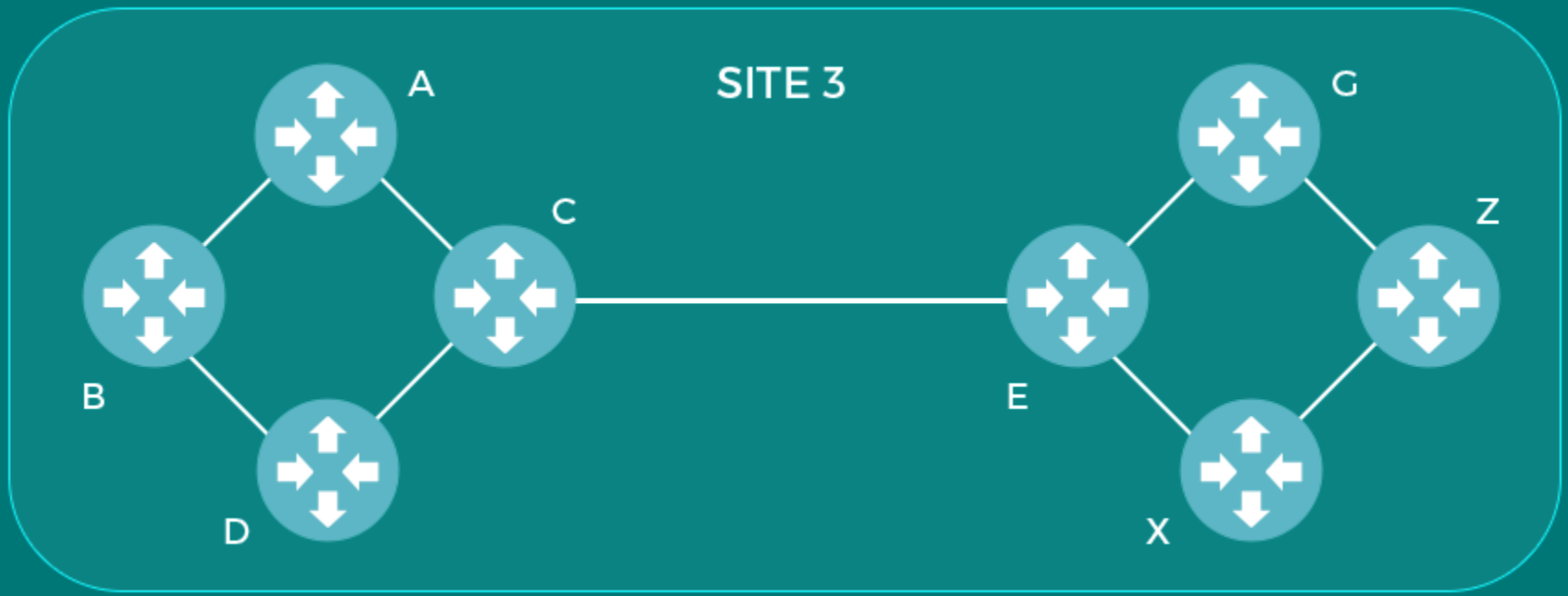
THE OBVIOUS SOLUTION



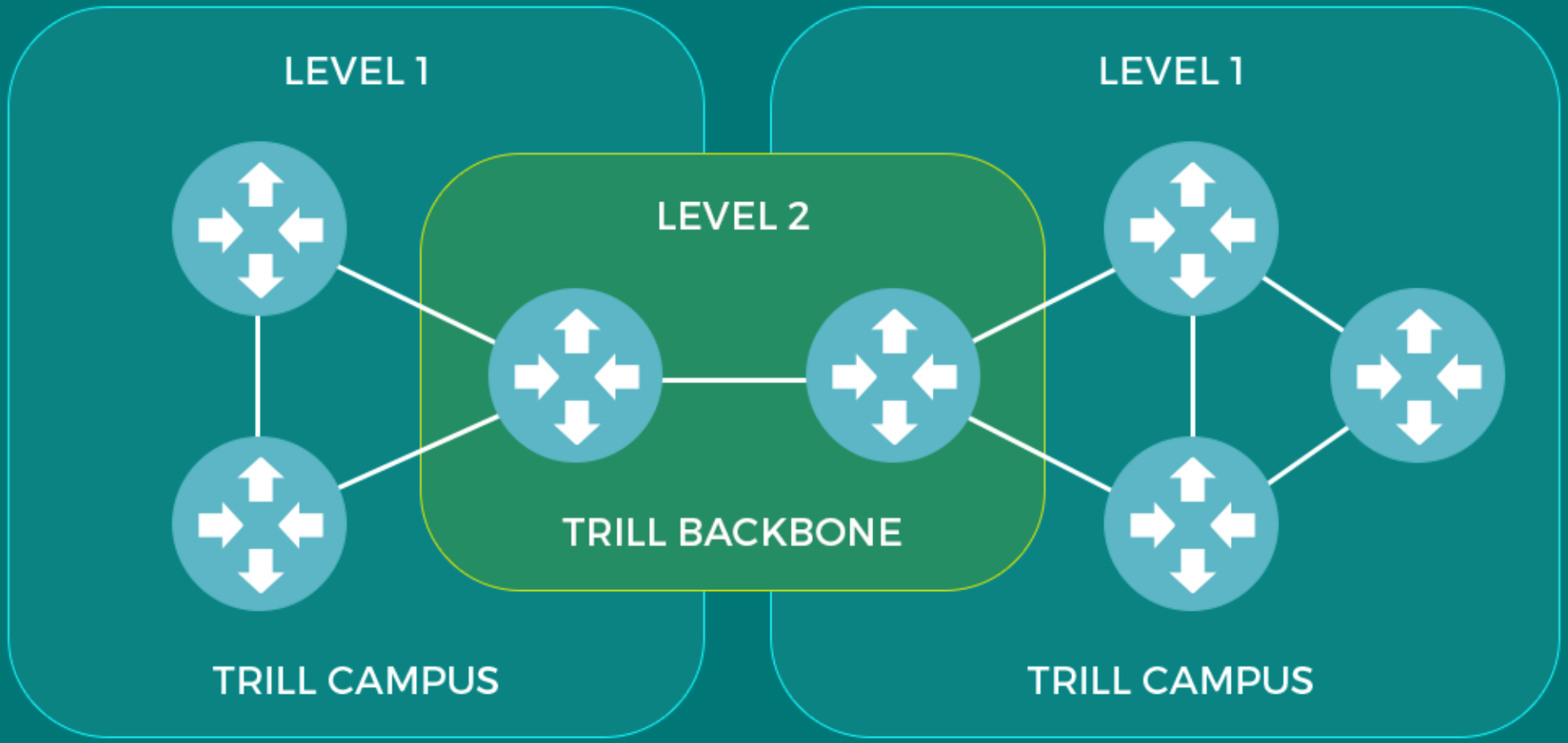
TRILL - ZONE EXTENSION?

- merge zone within trill:
  - topology calculation slow down
  - unique huge area to manage
  - all nicknames must be unique
  - multicast traffic non sense

MERGING ISSUES



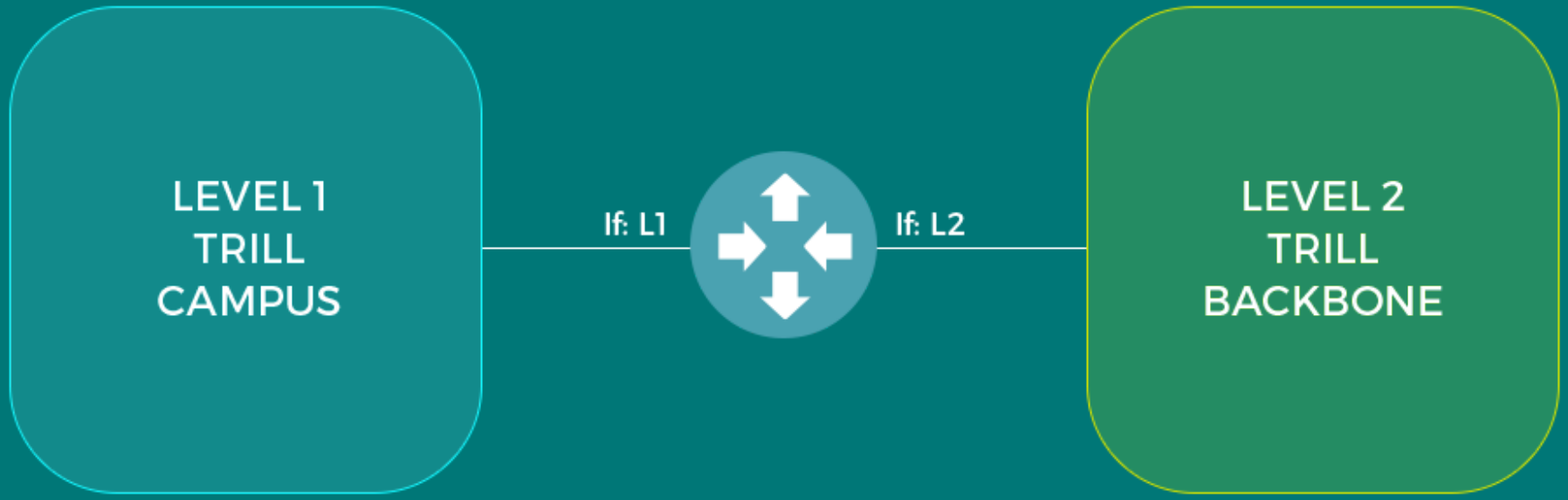
MULTI LEVEL TRILL PROTOCOL (MLTP)



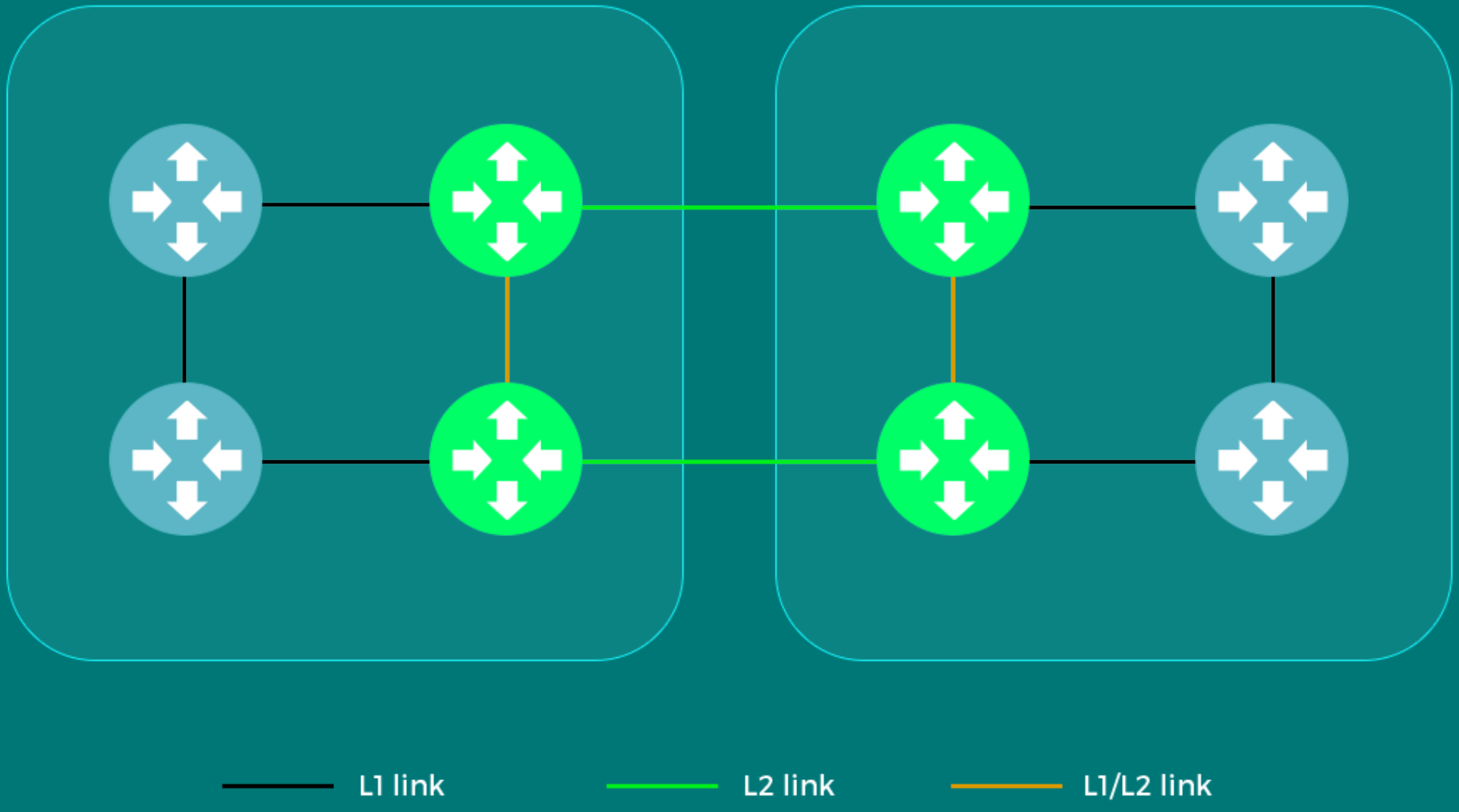
MLTP LEVELS

- level 1: campus
- level 2: backbone

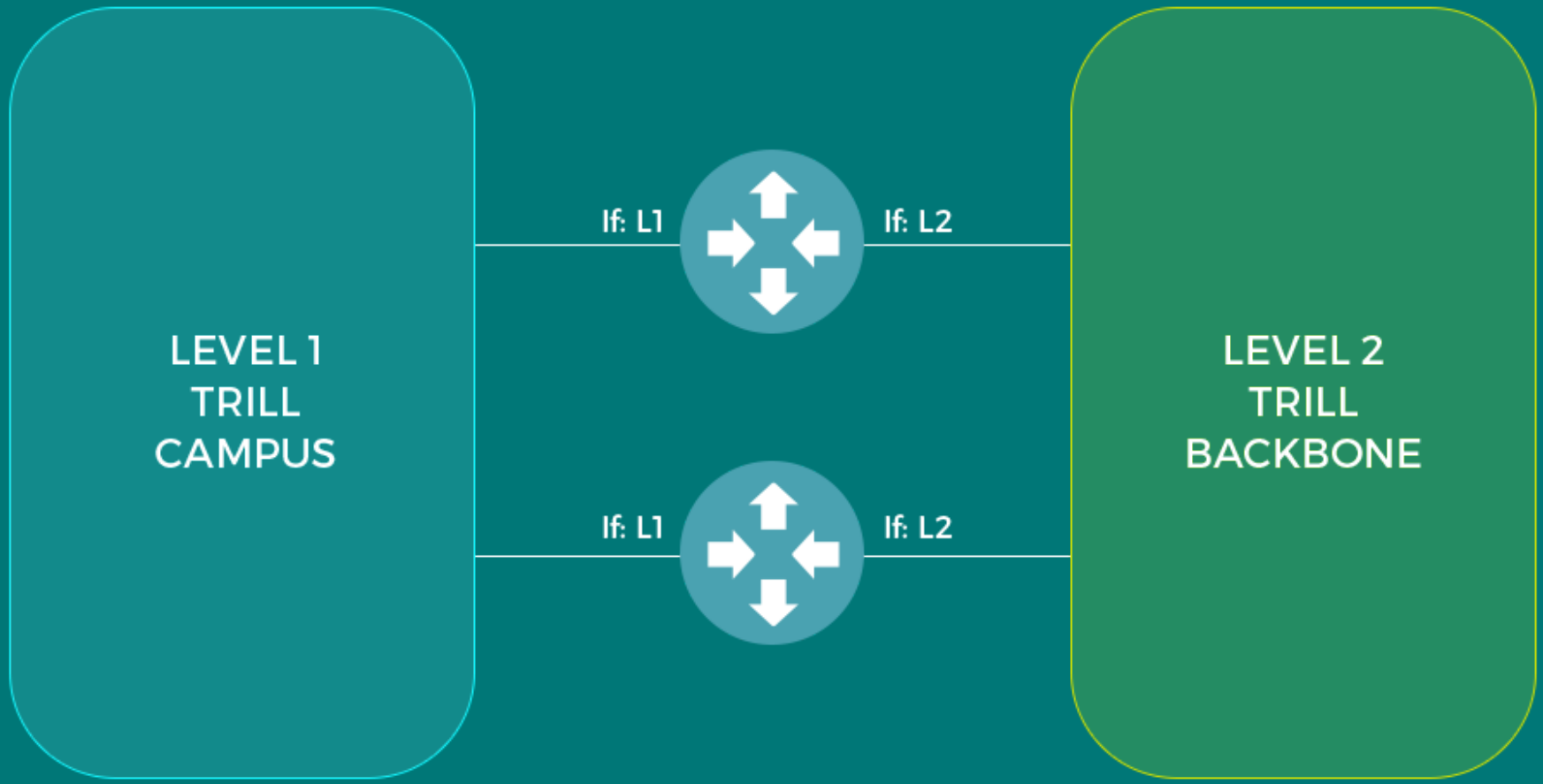
BORDER RBRIDGE



CONTROL PLANE: LINK LEVEL



GATEWAY REDUNDANCY

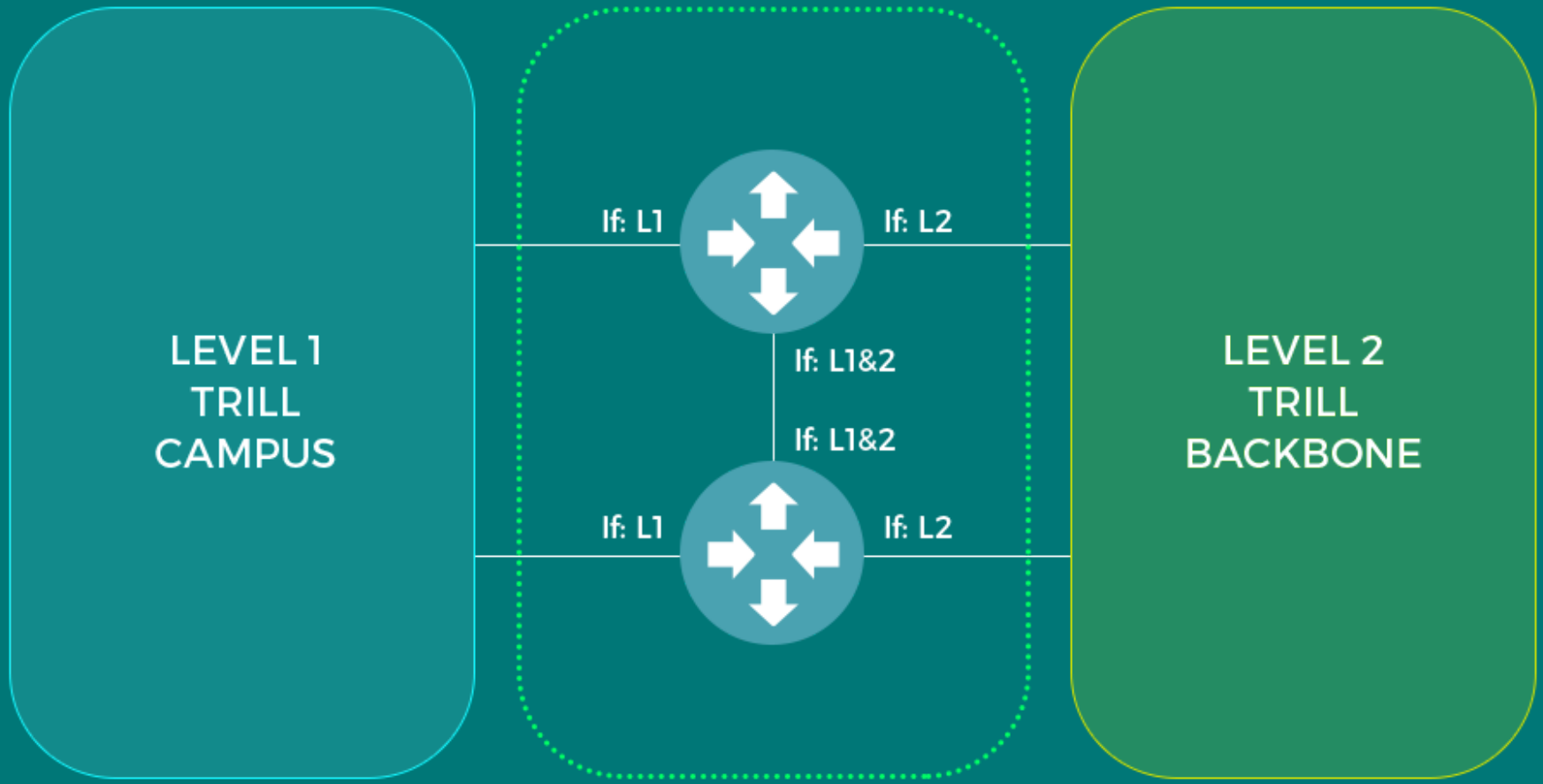


HAVING MANY GATEWAYS?

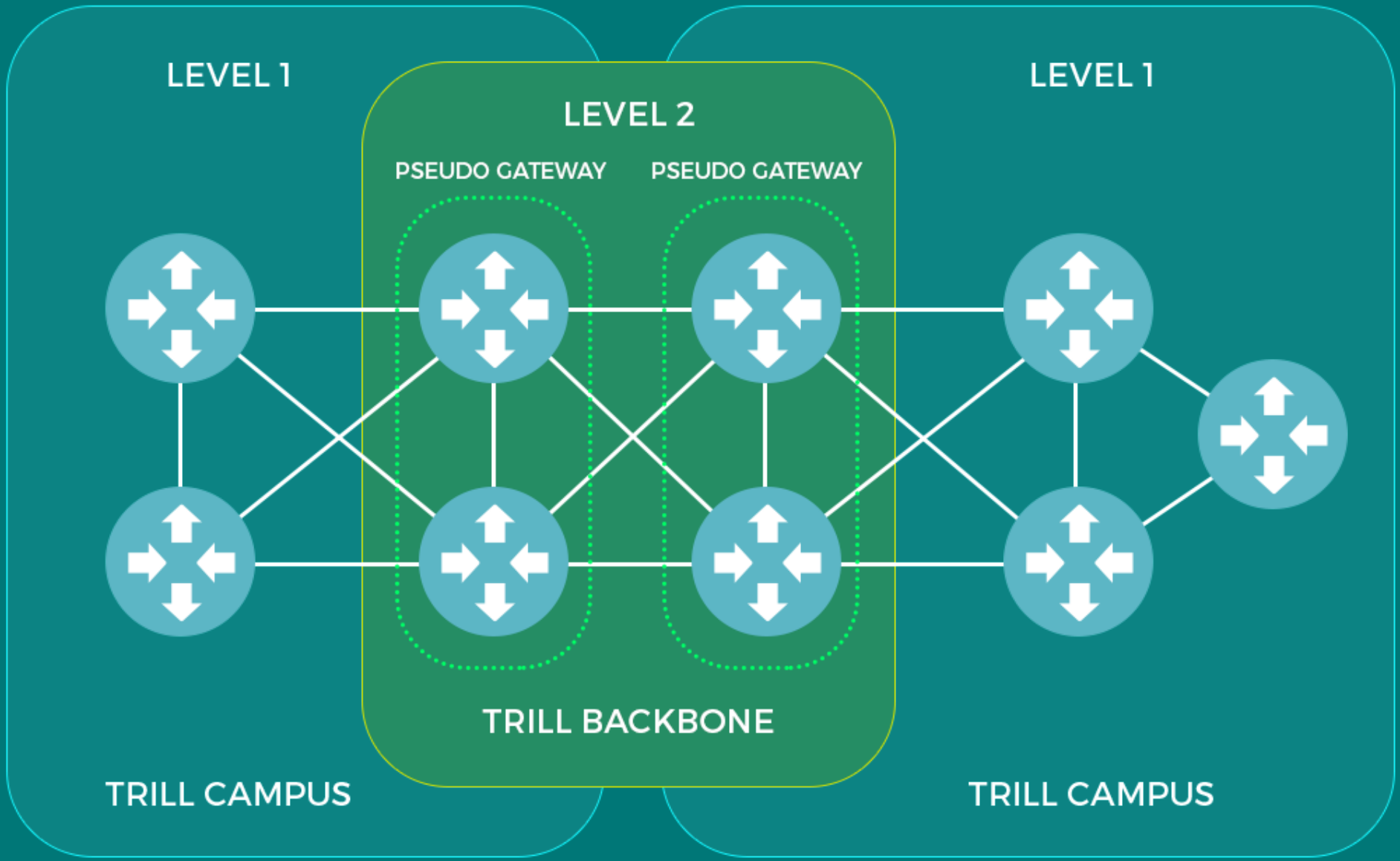
- choose the closest RBridge?
  - What if it fails?
  - What if the return path is different?

PSEUDO GATEWAY

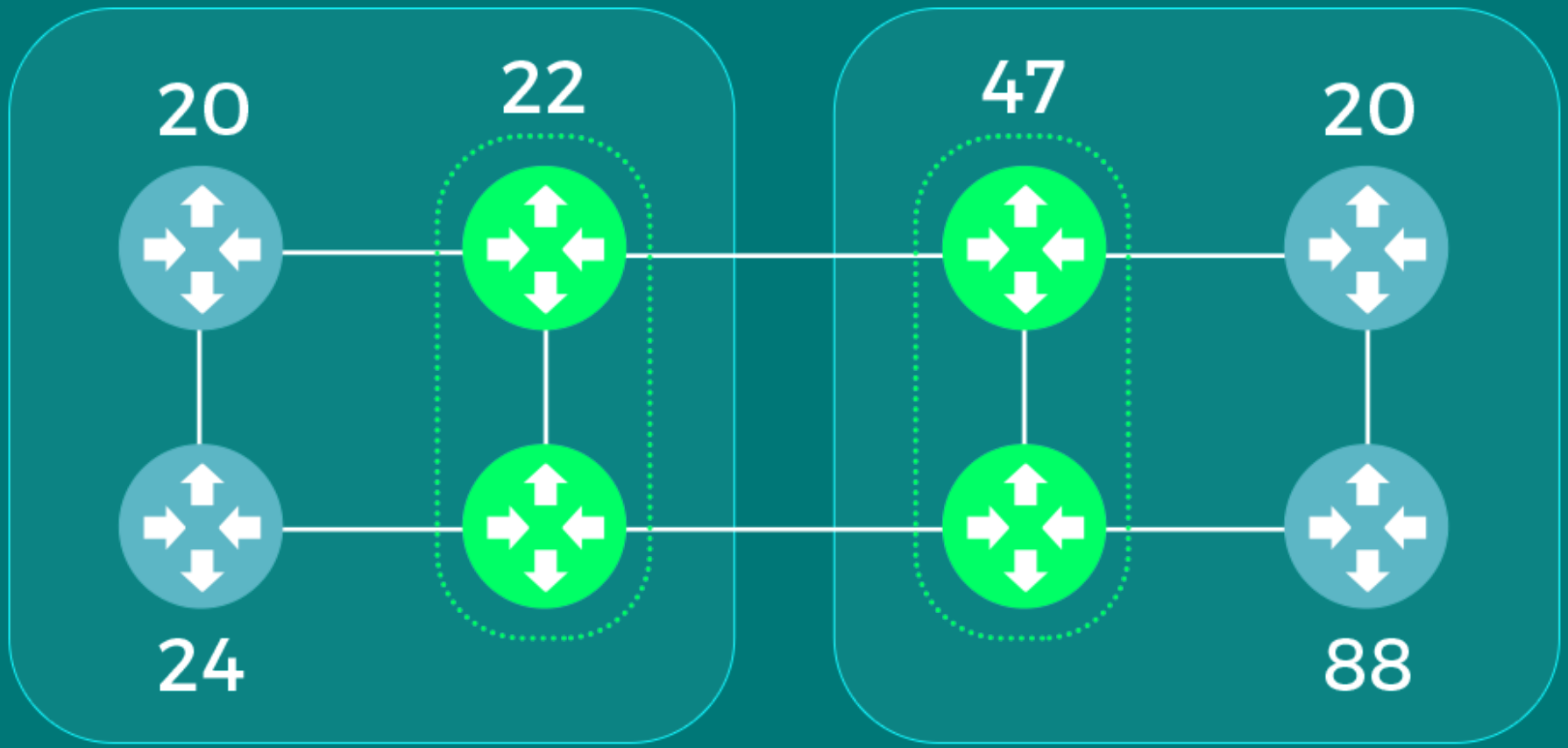
- Uniform way to go outside the campus
- Manage redundancy themselves

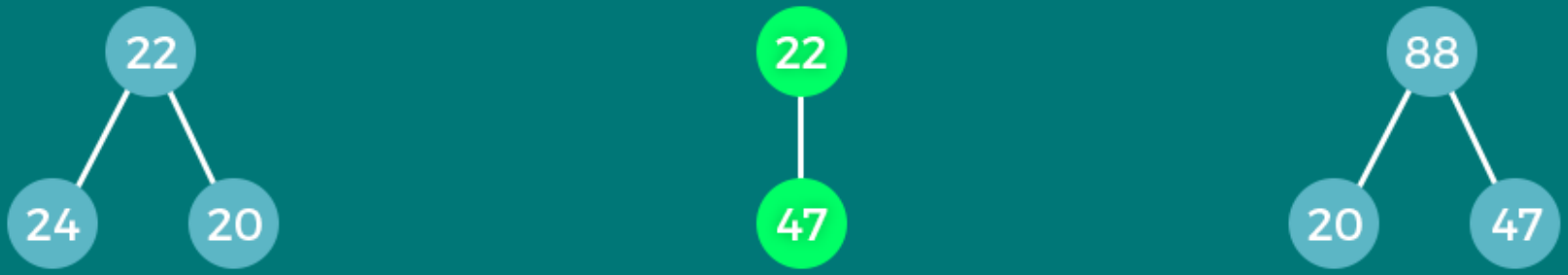

LEVEL 1
TRILL
CAMPUS
If: L1
If: L2
If: L1&2
If: L1&2
If: L1
If: L2
LEVEL 2
TRILL
BACKBONE

MULTI LEVEL TRILL PROTOCOL (MLTP)

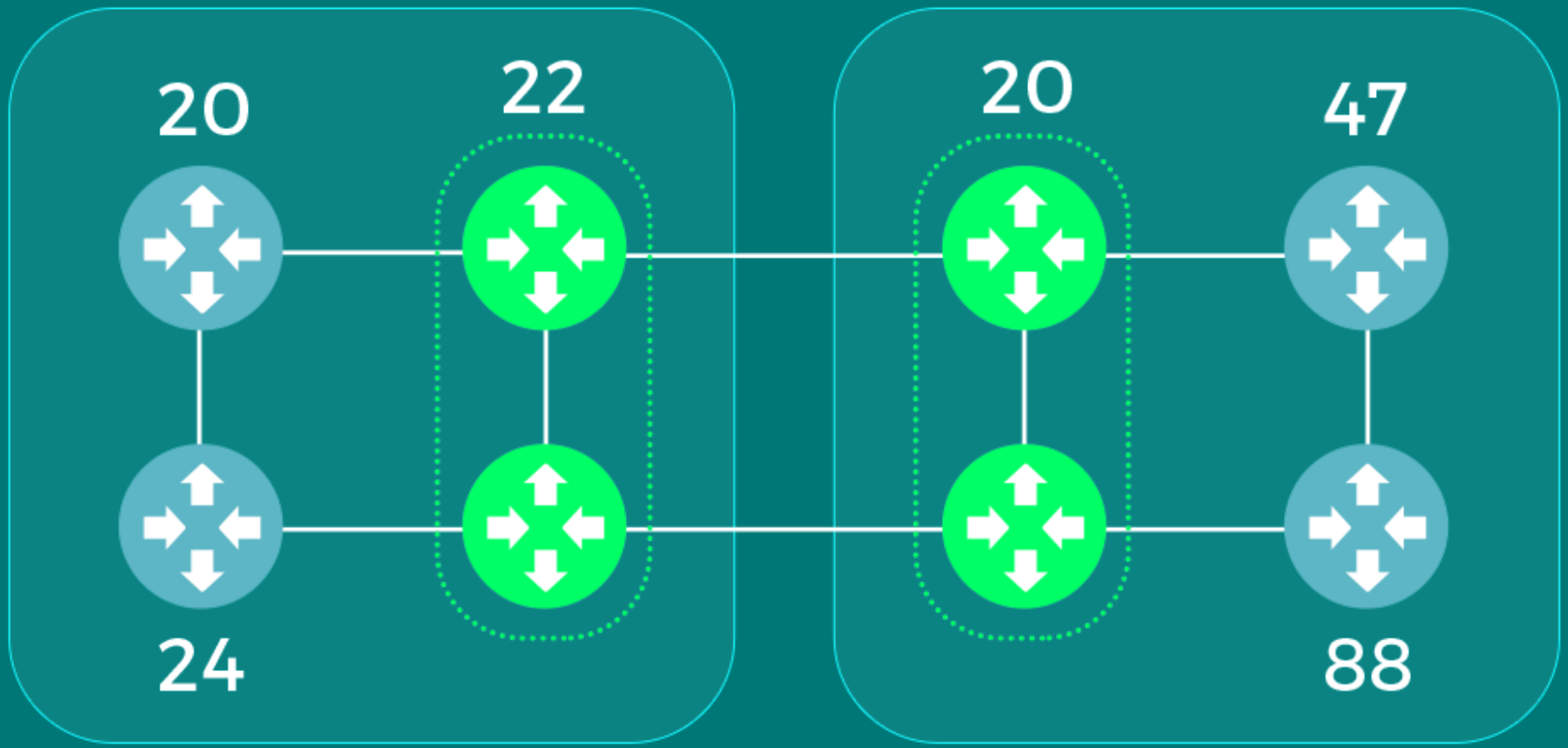


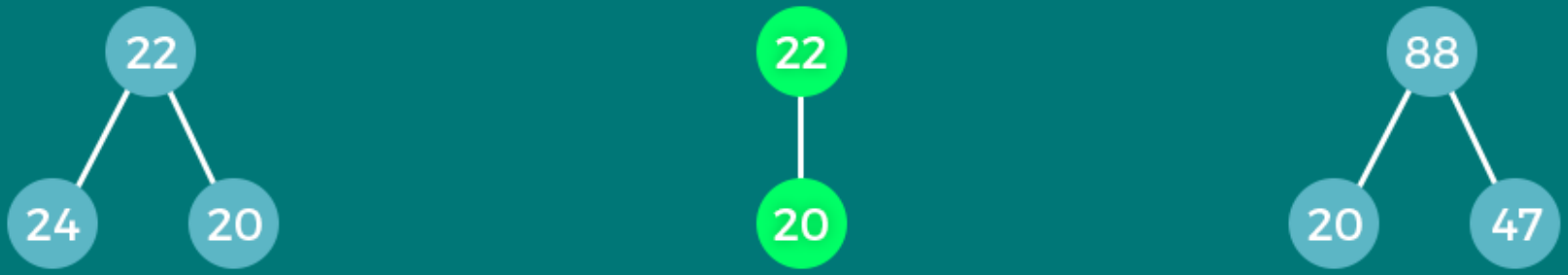
MULTI-LEVEL TOPOLOGY BUILDING



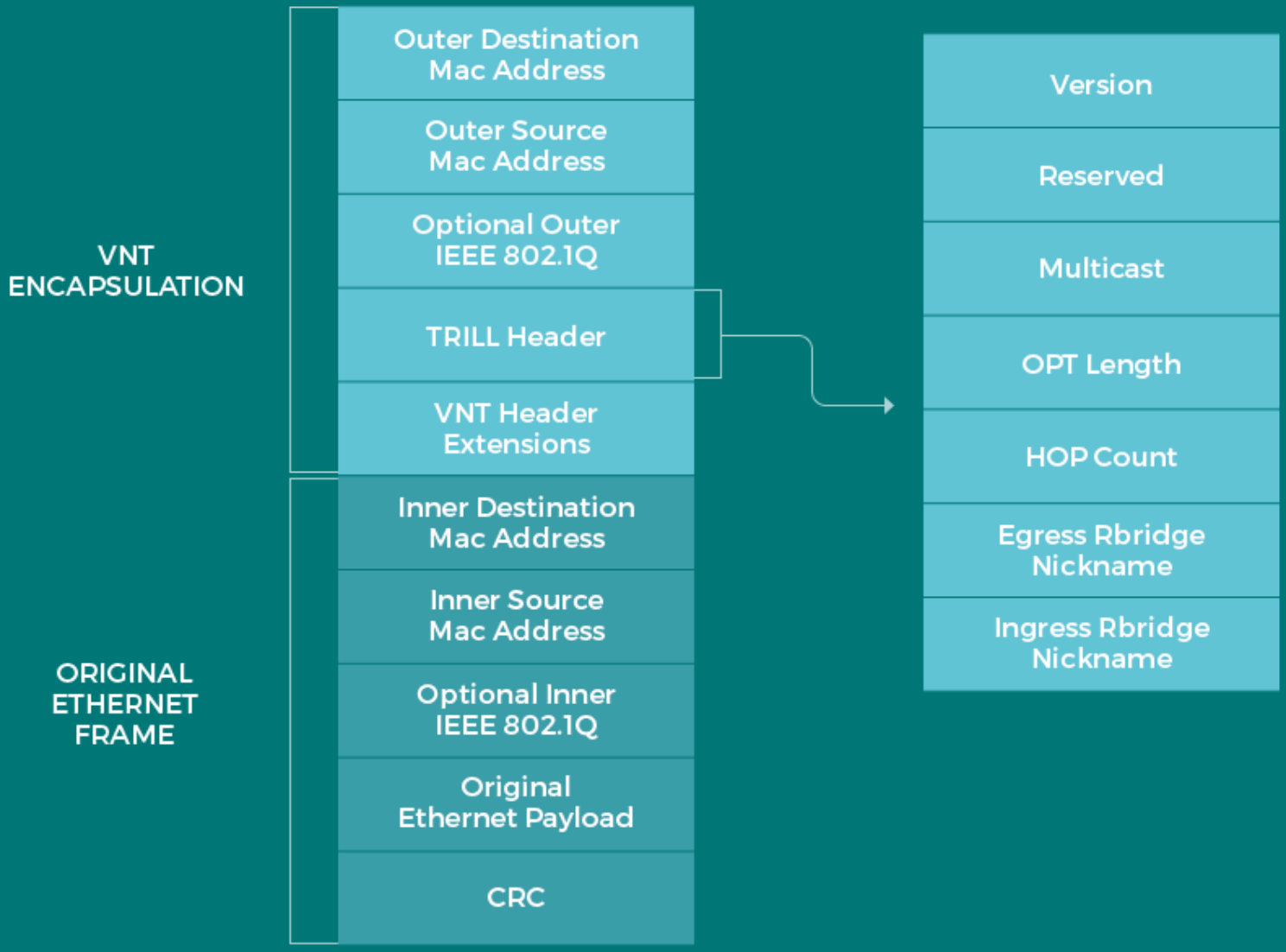


RESOLVING NICKNAME COLLISION

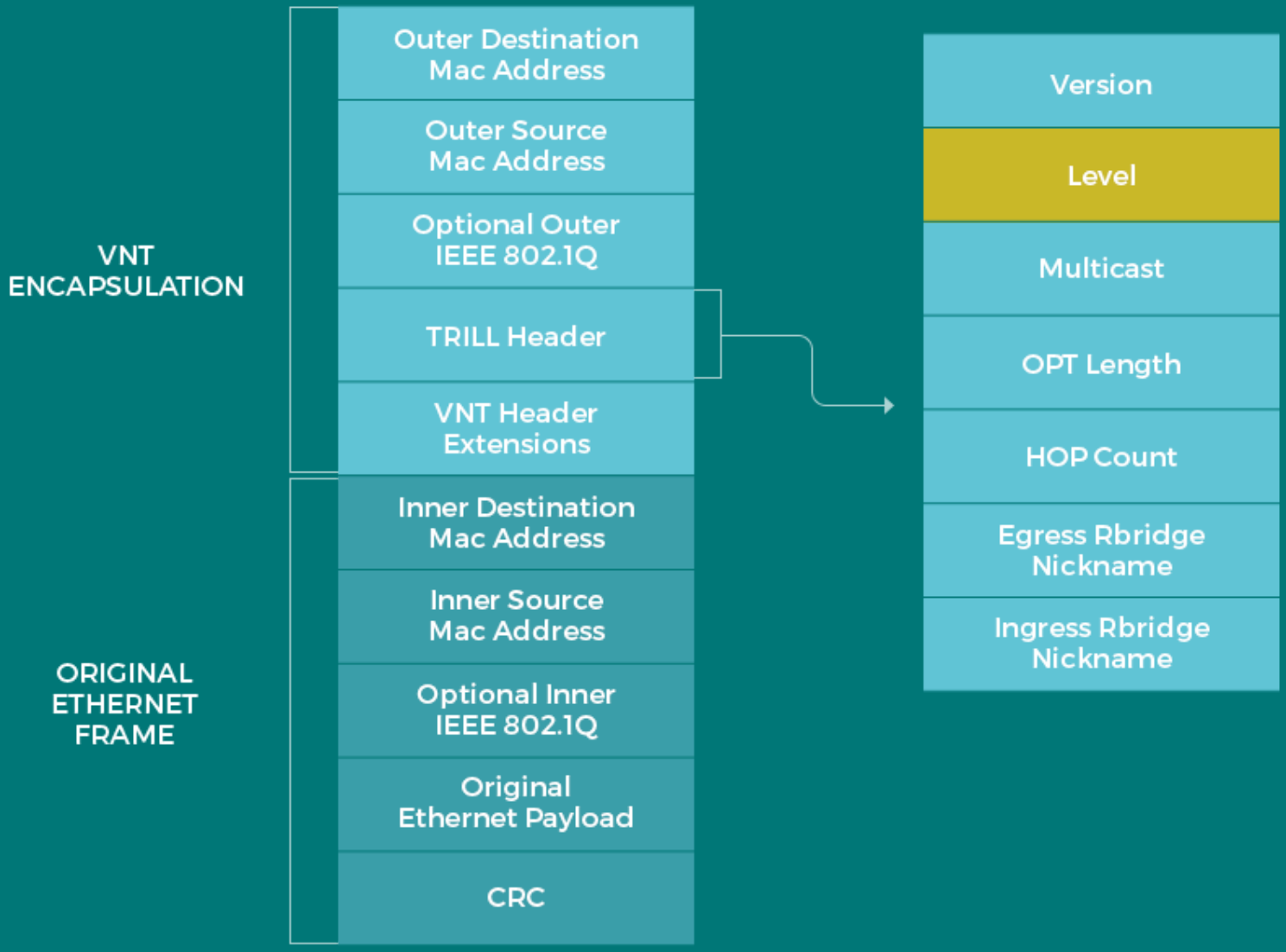




TRILL HEADER



MLTP HEADER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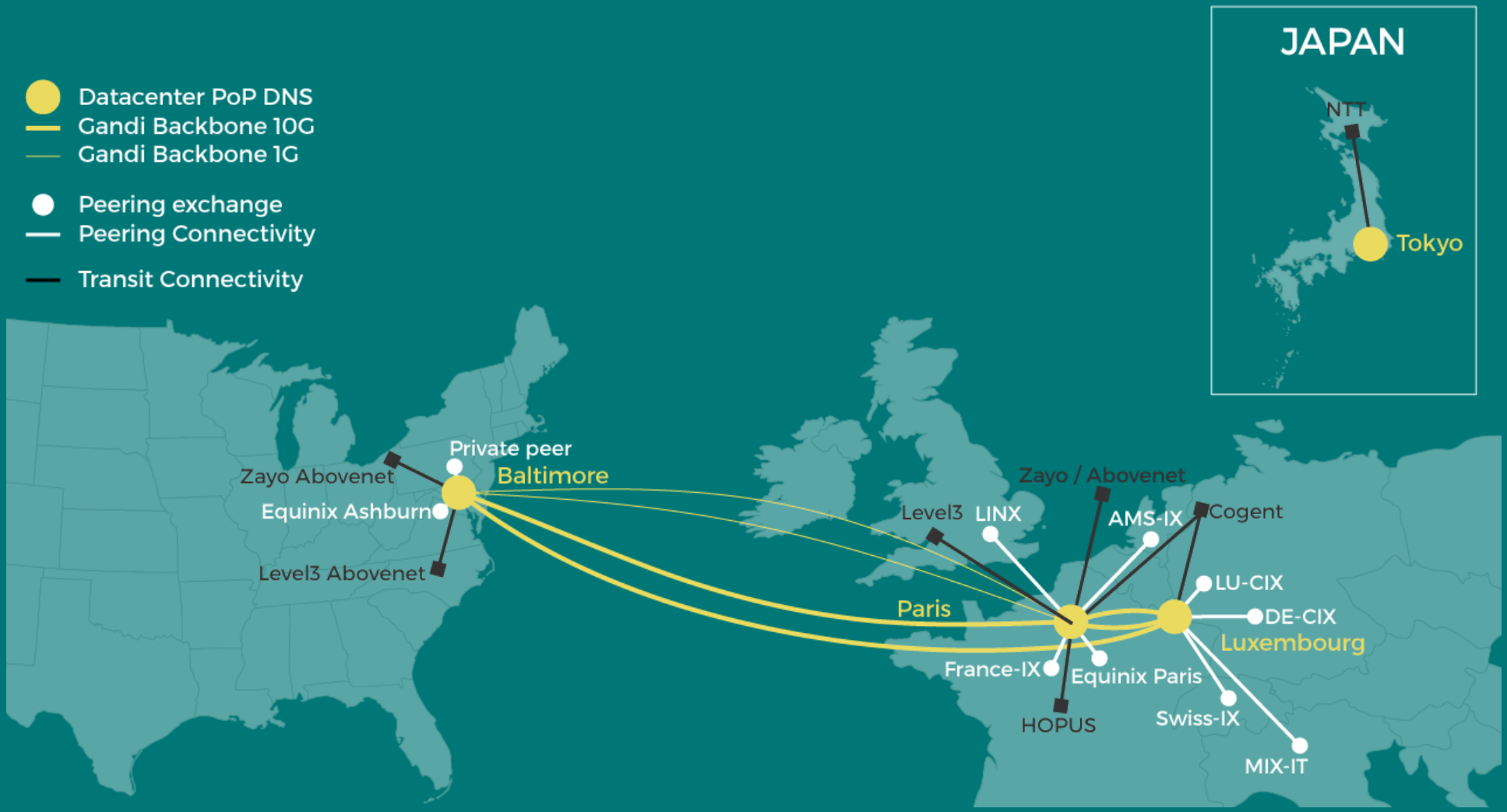
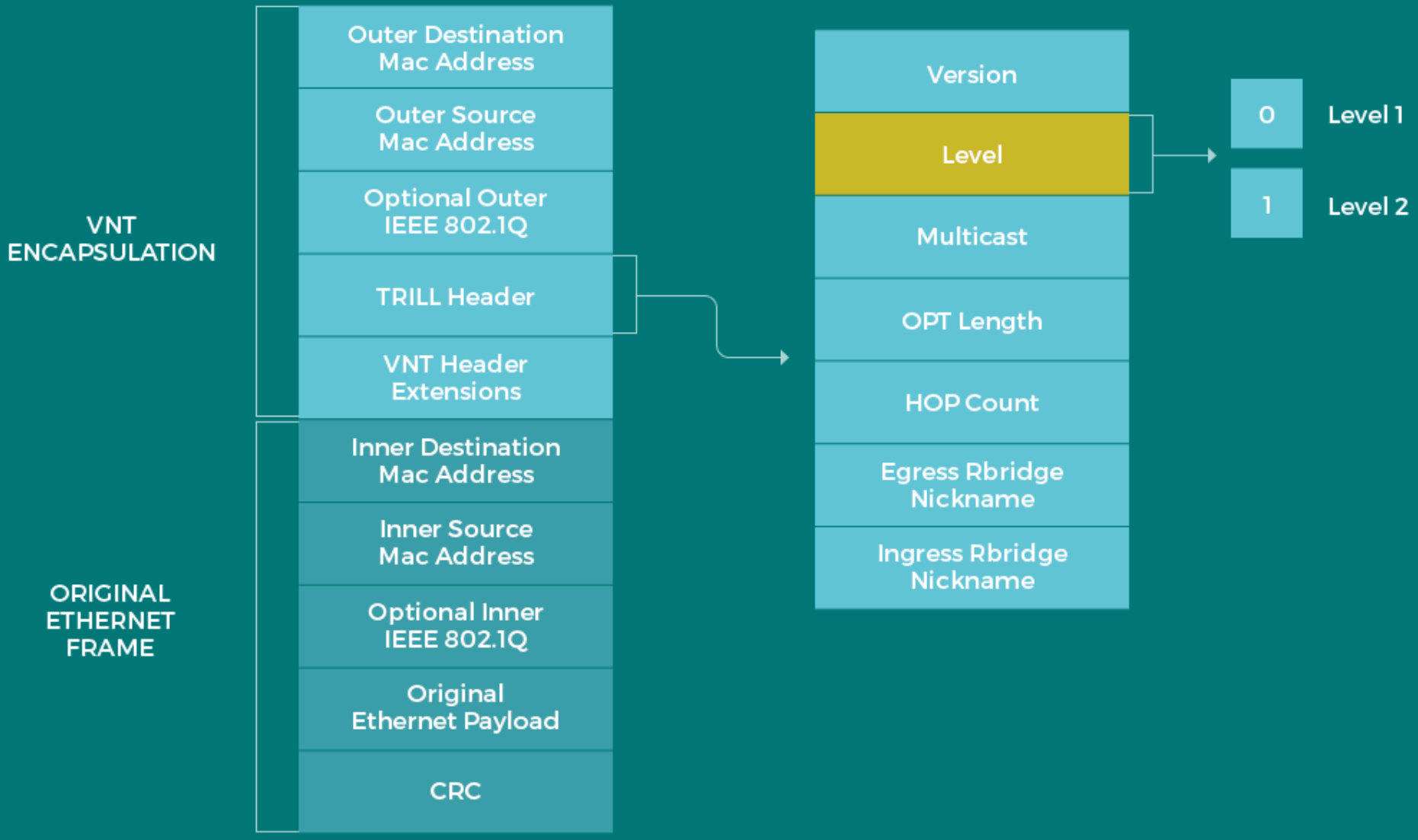
ORIGINAL ETHERNET FRAME
Inner Destination Mac Address
Inner Source Mac Address
Optional Inner IEEE 802.1Q
Original Ethernet Payload
CRC
Version
Level
Multicast
OPT Length
HOP Count
Egress Rbridge Nickname
Ingress Rbridge Nickname


Datacenter PoP DNS
Gandi Backbone 10G
Gandi Backbone 1G
Peering exchange
Peering Connectivity
Transit Connectivity
JAPAN
NTT
Tokyo
Private peer
Baltimore
Zayo Abovenet
Equinix Ashburn
Level3 Abovenet
Paris
Zayo / Abovenet
Level3
LINX
AMS-IX
Cogent
LU-CIX
DE-CIX
Luxembourg
France-IX
Equinix Paris
HOPUS
Swiss-IX
MIX-IT

MLTP LEVEL VALUES



TRILL BASE CODE EVOLUTION

- first implementation in Linux Bridge
  - version evolution in production (v3.4 - v3.10 - v3.14 - v4.1)
- testing public network using TRILL

FUTURE DEVELOPMENT?

- Control plane: OpenVSwitch
  - more attractive nowadays
  - central view
- Data plane: programmable network cards
  - move the dataplane closer to the network to reduce latency

IEEE CLOUDNET 2015

Valentin Del Piccolo (Ph.d student), William Dauchy

"Multi-Tenant Isolation in a TRILL Based Multi-Campus Network"

Gandi SAS - University Pierre & Marie Curie

TRILL + VNT sources: github.com/Gandi/ktrill

GANDI.NET

Gandi Hosting - gandi.net/hosting

William Dauchy - william@gandi.net

slides pres.gandi.net/kr2015